

How V-JEPA 2 represents physical variables in its latent space

Direction, **speed** and **acceleration** of a moving disk: probing, nullspace, steering, and manifold (spline) steering

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Four findings at a glance

What a frozen V-JEPA 2 ViT-L/16 encodes about a moving disk's direction, speed and acceleration

1

Readable from block 1

All three variables are linearly readable after the first transformer block.

[evidence from slide 6 →](#)

2

Counts measure the probe

The nullspace “dimension count” measures the probe procedure, not the representation.

[evidence from slide 8 →](#)

3

Edits fade downstream

Steering edits work at the edited layer and fade within a few blocks, for probe and spline steering alike.

[evidence from slide 10 →](#)

4

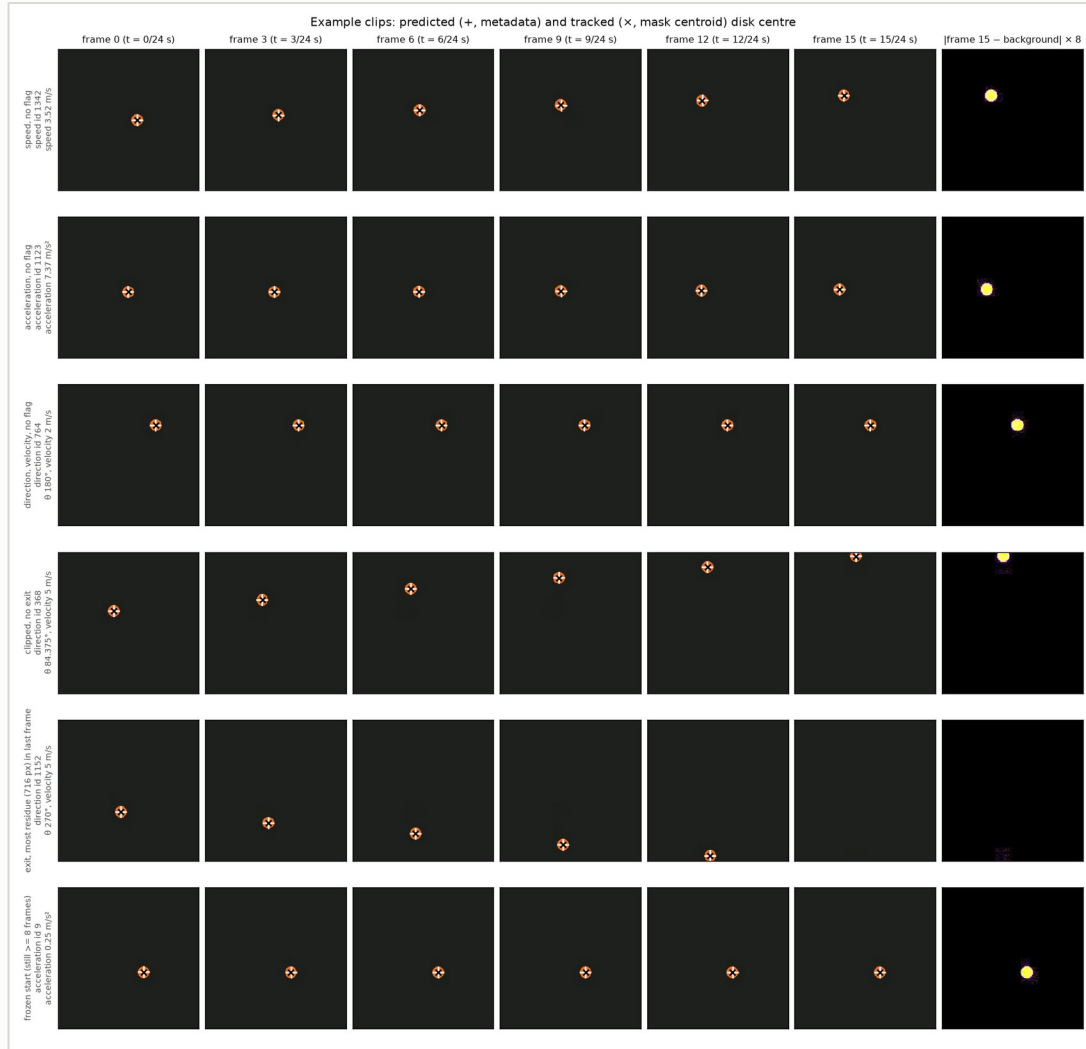
Curved, but no fix

The representation is a curved, low-dimensional manifold; following it keeps paths natural but doesn't make edits propagate.

[evidence from slide 12 →](#)

The data audit shaped what we can claim

4,572 synthetic clips, 16 frames at 24 fps, 256×256 ; every clip decoded, tracked and checked against its metadata



Three sets: direction 1,500 clips (64 angles; half constant velocity, half accelerating from rest), speed 1,536, acceleration 1,536

The audit found: an orange disk (DATA.md: blue); manifests sorted by label; 113 direction clips where the disk leaves the frame

The confound: speed and acceleration labels are exactly proportional to distance travelled ($0.625 \times \text{speed}$, $0.1953 \times \text{acceleration}$)

Protocol: frozen encoder, own hooks, held-out values

How representations are extracted, pooled and kept apart from evaluation

EXTRACTION



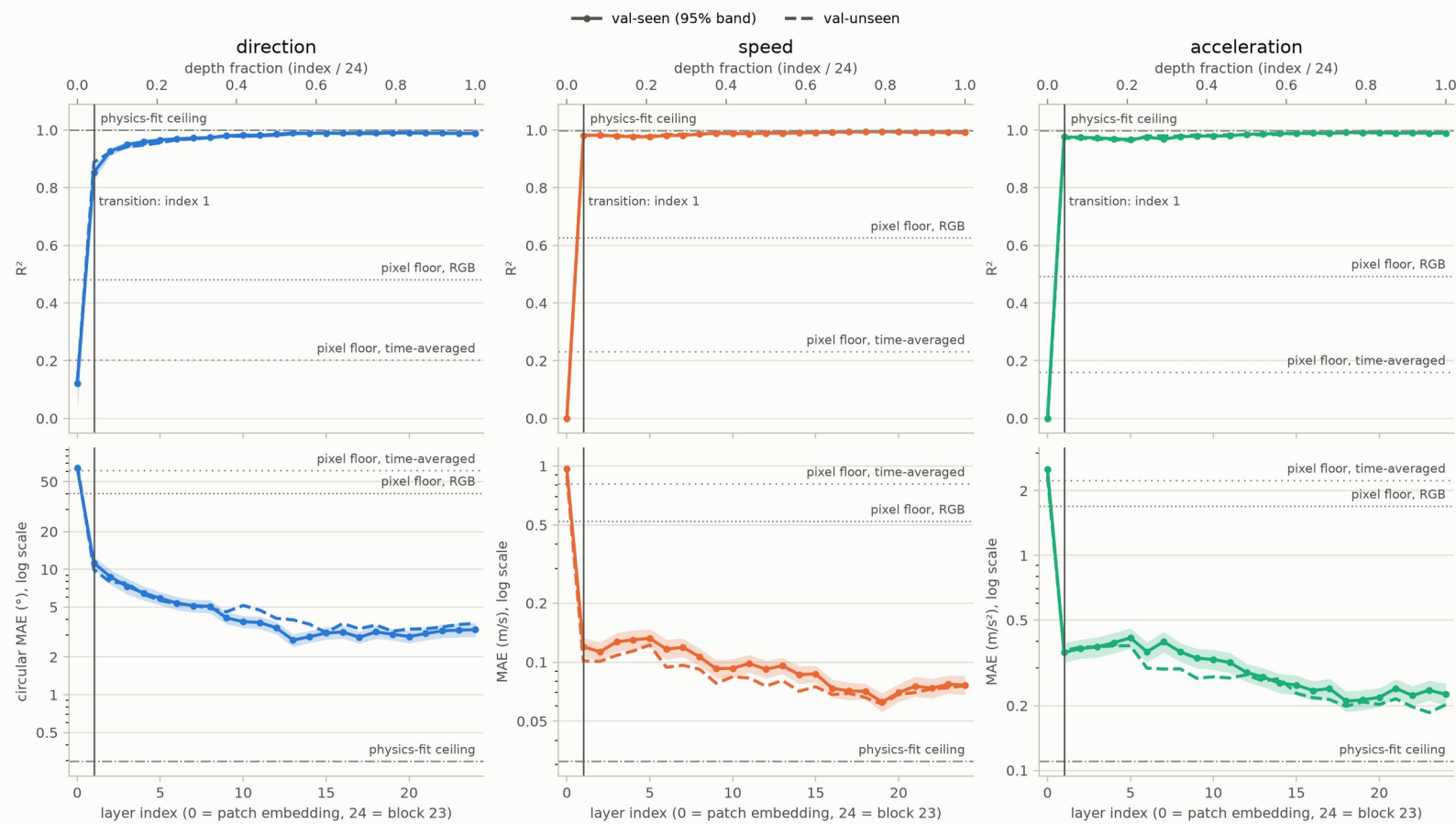
SPLITS · clips per role (direction / speed / acceleration); 12 of 64 values per set never appear in training

train	val-seen	val-unseen	test-seen	test-unseen
813 / 832 / 832	203 / 208 / 208	94 / 96 / 96	203 / 208 / 208	187 / 192 / 192
fit probes, nullspace rounds, curves	layer and model choices	4 values never in train; held-out check	read once; 15 steered clips per variable	8 values never in train; read once; 15 steered clips per variable

Fit on train · choose on validation (steering readouts are fit on both validation roles) · evaluate on test once, from committed code. Speed and acceleration share one assignment by clip id; direction is stratified by motion group × 45° octant.

Motion is linearly readable after the first block

Metric: R^2 of a ridge probe on the mean activation, validation clips (95 % bands); held-out values dashed



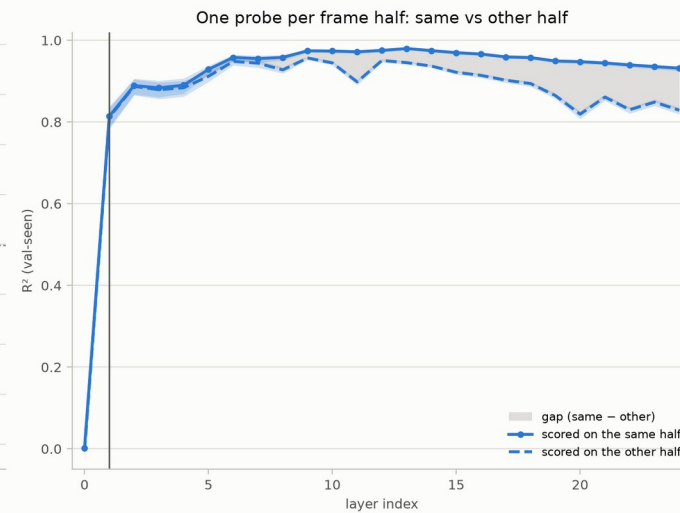
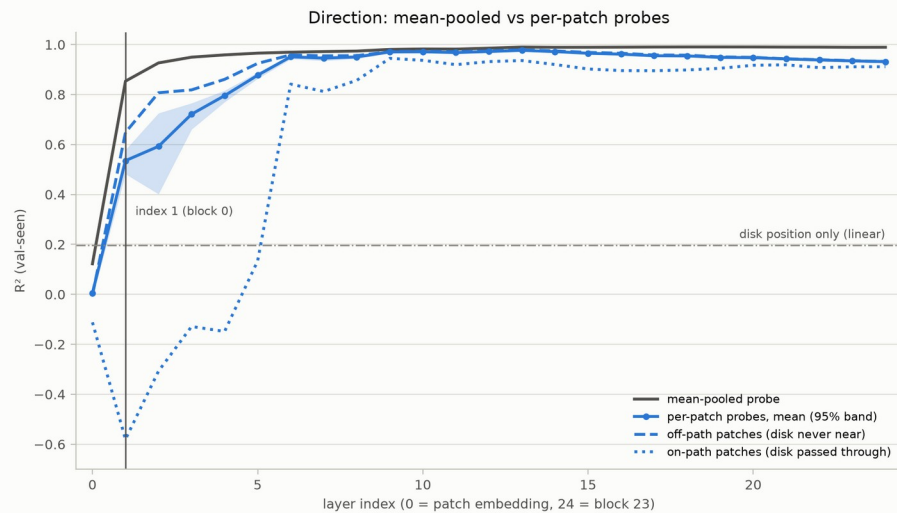
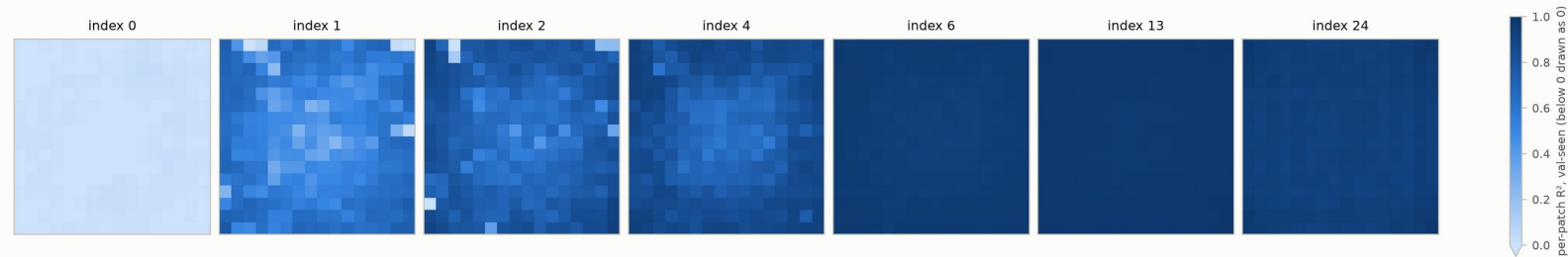
Readable at index 1 for all three (R^2 0.85 / 0.98 / 0.98; transition at index 1 in all 10,000 bootstrap resamples); ≈ 0.99 by index 17–19; test agrees

Far above pixels (raw RGB 0.48–0.63), below a physics fit (≥ 0.997)

Speed and acceleration early, as in the paper; direction readable earlier than the paper's emergence zone

Direction is spread across the frame from block 1

Metric: R^2 of per-patch direction probes (one probe per patch position, time-averaged), validation clips



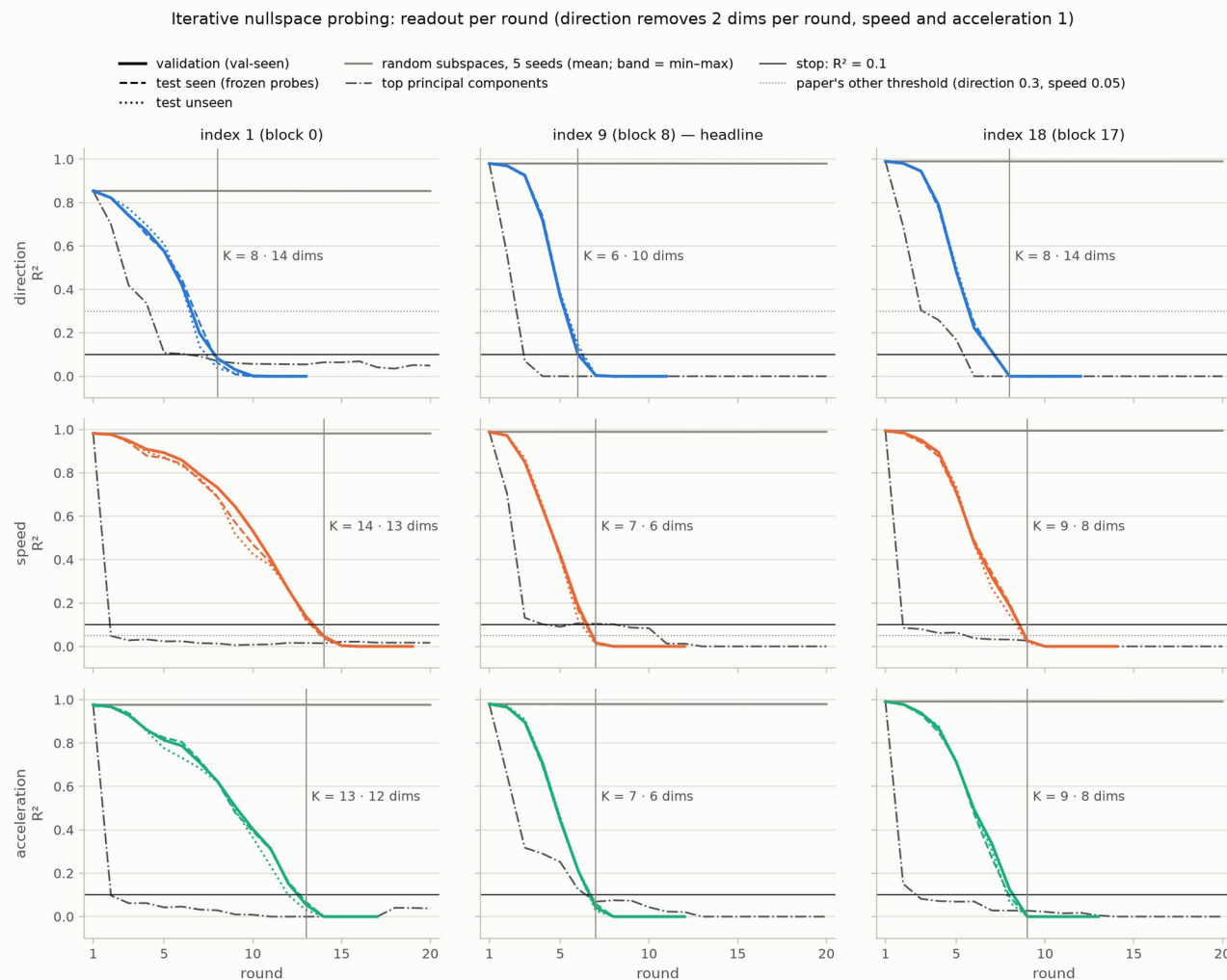
Patches the disk never came near: R^2 0.65 at index 1, 0.96 at index 6; disk position alone explains only 0.20

One probe generalizes across the frame: trained on one half, it reads the other half (gap 0.003 at index 1); position-specific coding appears only late (gap 0.13 at index 20)

The paper places this local-to-global shift at one-third depth; here it happens after one block

A handful of rounds erase the linear readout

Fit a probe, remove its directions, refit. Metric: validation R^2 per round; K = first round with $R^2 < 0.1$



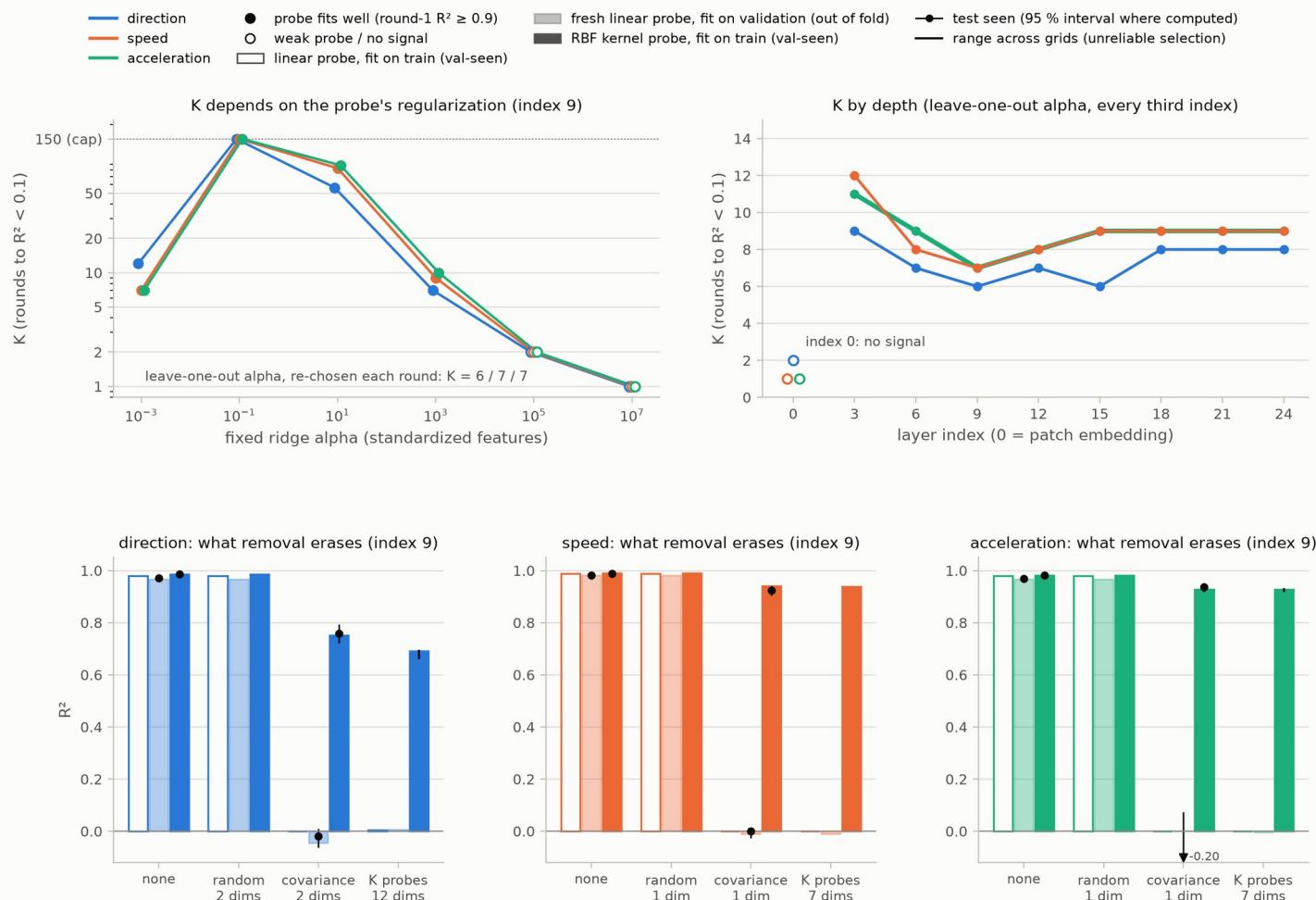
$K = 6 / 7 / 7$ at index 9; 6–14 rounds at indices 1, 9 and 18; the readout is gone after removing 6–14 dimensions; test agrees within one round

Random directions change nothing (R^2 drops ≤ 0.014 even after 150 rounds); top principal components erase it as fast or faster

No growth with depth: K stays at 6–12 rounds from index 3 to 24

The nullspace count measures the procedure, not the representation

Index 9. Top: K under fixed ridge penalties. Bottom: R^2 after removal, linear vs RBF kernel probe



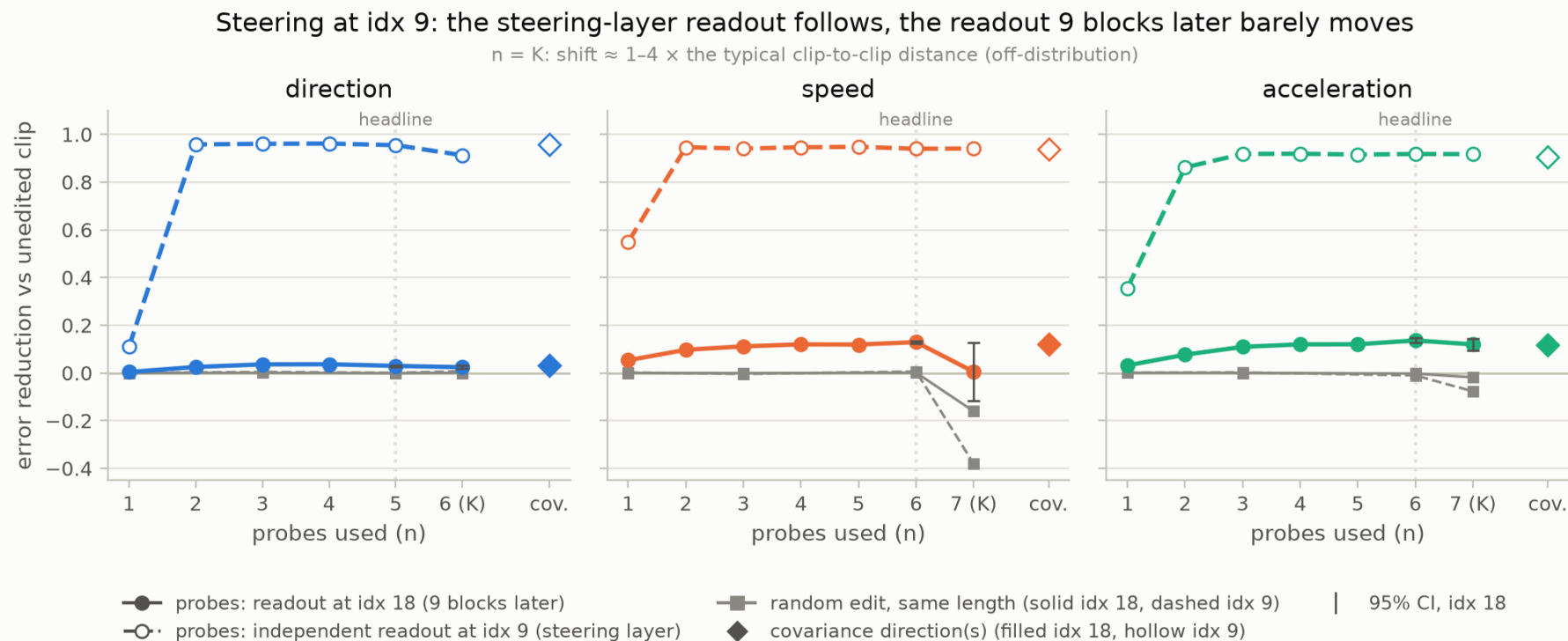
Same layer and data, only the ridge penalty changes: K moves from 7 to more than 150 among probes that read well (round-1 $R^2 \geq 0.93$)

Removing only the label-covariance direction (1 dimension; 2 for direction) takes a fresh linear probe to chance: a property of linear regression

An RBF kernel still reads each variable after erasure: test 95 % CIs between 0.72 and 0.95; shuffled labels ≤ 0.005

Steering works at the steering layer

Metric: error reduction vs the unedited clip (1 = reaches the target, 0 = no change). Edit at index 9 on 30 test clips × 5 targets, 3 never seen in training



An independent readout at the steering layer follows the edit:

0.92–0.95 with $K - 1$ probes; random edits of the same length ≈ 0

One probe is not enough, even here

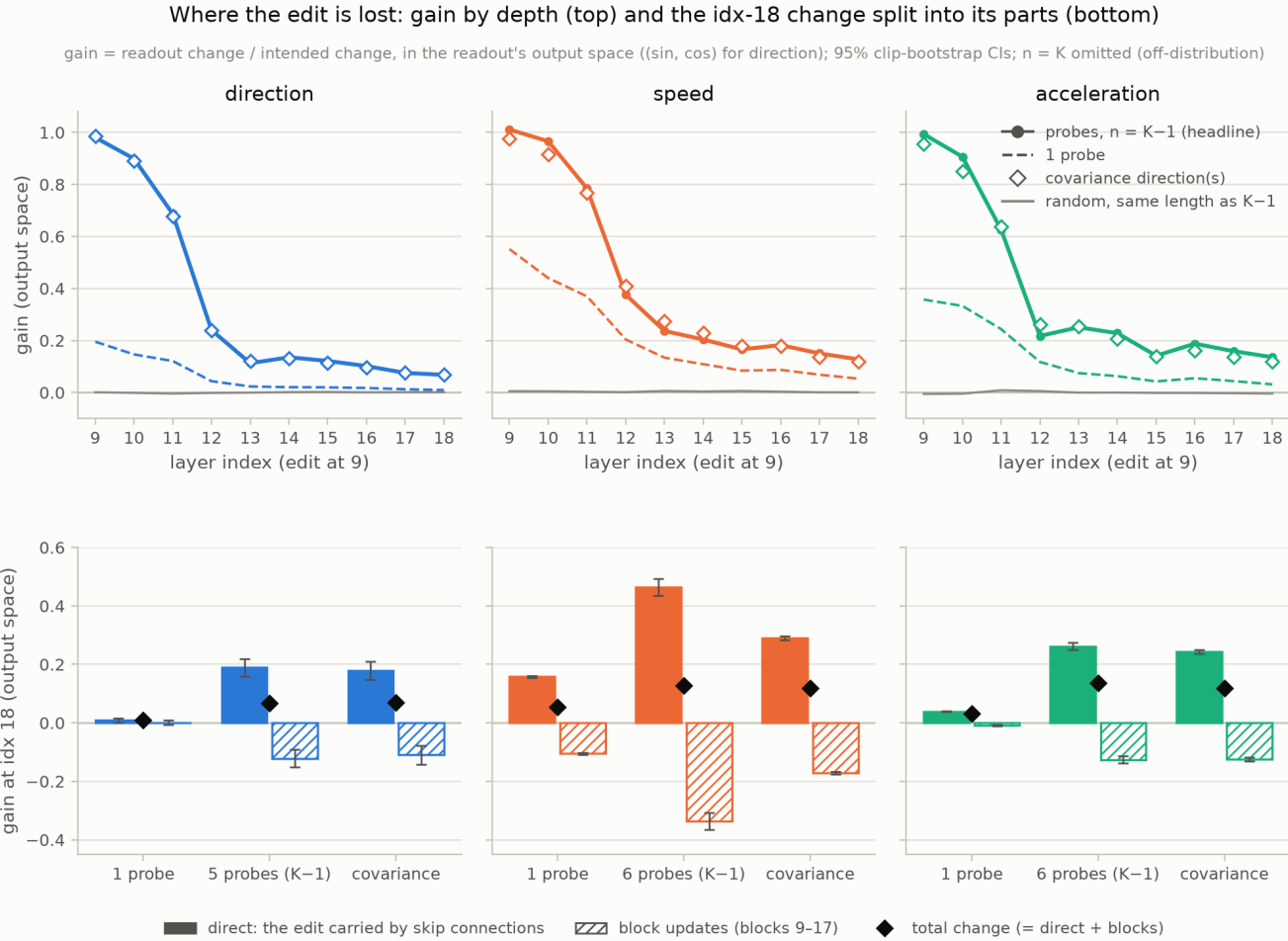
(0.11 / 0.55 / 0.36); the covariance direction alone does as well as $K - 1$ probes

Nine blocks later (index 18):

only 0.03 / 0.13 / 0.14

The edit is lost within three blocks

Metric: output-space gain = readout change ÷ intended change (1 = full effect). Top: by depth. Bottom: the index-18 change split into its parts



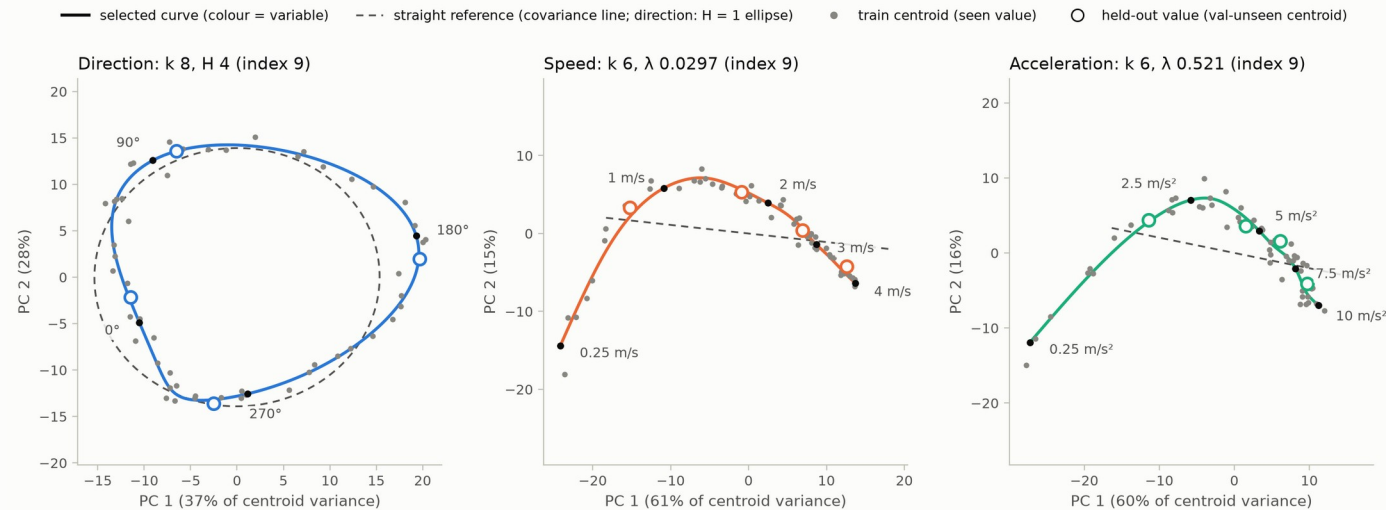
Gain 0.98–1.01 at index 9 → 0.22–0.38 at index 12 → 0.07–0.14 at index 18; random ≤ 0.03

The edit carried unchanged by the skip connections would move the index-18 readout by only 0.19 / 0.46 / 0.26: the later readout is only partly aligned with it

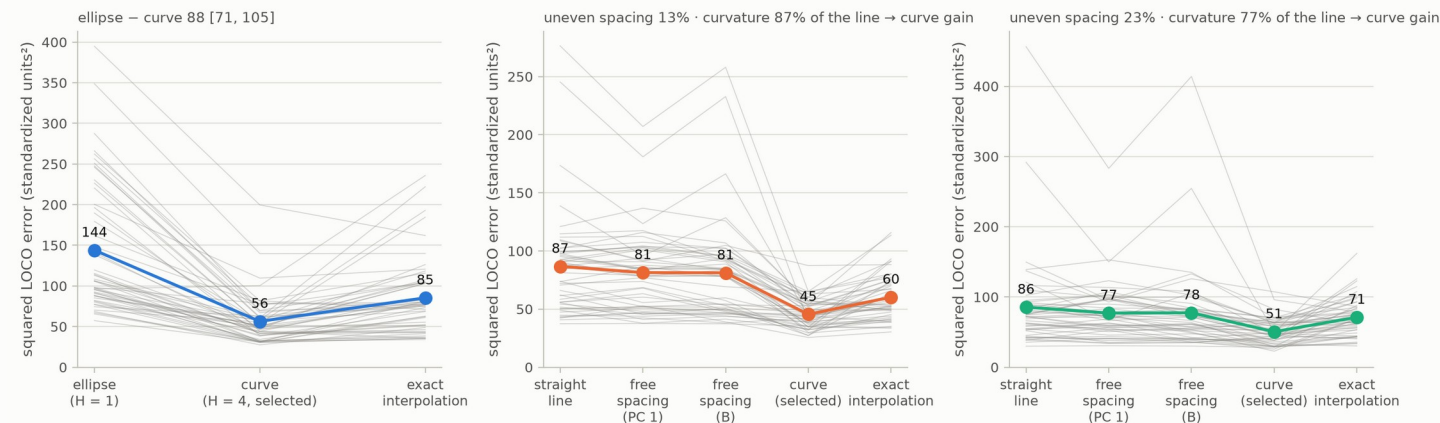
Block updates then push back 64 / 73 / 48 % of that

Part 2 · Curved, low-dimensional manifolds; direction is a loop

Index 9, per-value training centroids in PCA space. Top: fitted curves, held-out values as open circles. Bottom: held-out centroid error by curve family



Held-out centroid error by curve family (leave one centroid out; grey = one seen value)



Construction: per-value training centroids → smoothing spline (speed, acceleration) or closed trig polynomial (direction); dimension and smoothing chosen by leaving one value out

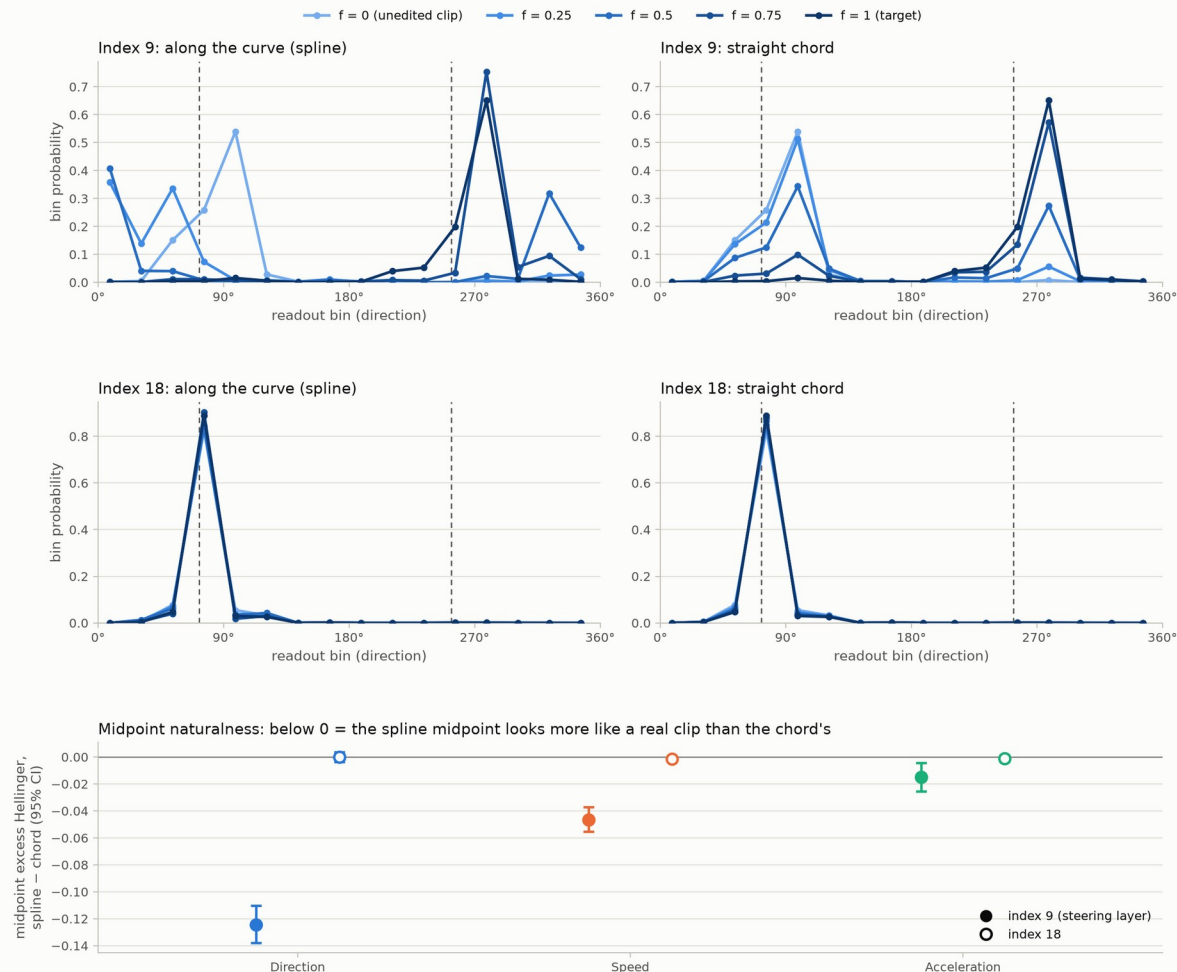
Held-out centroid error, simplest reference → curve: 144 → 56 (ellipse) / 87 → 45 / 86 → 51 (line); the curve wins at 12 / 12 held-out values

Low-dimensional: 6–8 PCA dimensions; Goodfire-style exact interpolation is 1.3–1.5 × worse

Part 2 · Along the curve, paths stay natural at the steering layer

One direction clip steered about 180° (72° → 253.1°), 16-bin readout along the path. Bottom: midpoint naturalness, spline – chord (below 0 = spline more natural)

Direction, clip 318: start 72° → target 253.1° (dashed lines; largest start-target angle among the 150 pairs)



At the steering layer, the straight chord's midpoint splits between start and target (bimodal in 47 % of far-target cases); spline midpoints stay closer to real clips (spline – chord -0.124 for direction)

Held-out design: the 90 test clips from Part 1 (30 per variable), 5 targets (3 with no centroid in the fit), readouts fit on validation clips, read at every index 9–18

At index 18 every difference vanishes: every edit still reads the start

Part 2 · Downstream, every method collapses alike

Metric: output-space gain (1 = full effect), edit at index 9. Top: gain by depth for four methods. Bottom: time-structured edit minus uniform edit

Propagation of the edit from the steering layer (index 9) to index 18; colour = variable



Spline, covariance line and probes all fall to 0.07–0.14 at index 18; spline and covariance within 0.005

The spline edit points 66–76 % differently from the covariance edit, yet propagates identically

Only timing nudges it: a per-time-step edit adds +0.01–0.03 and reversed timing hurts; below the pre-declared +0.05 threshold, so not confirmed

Spline vs multi-probe: strengths, limitations, failure cases

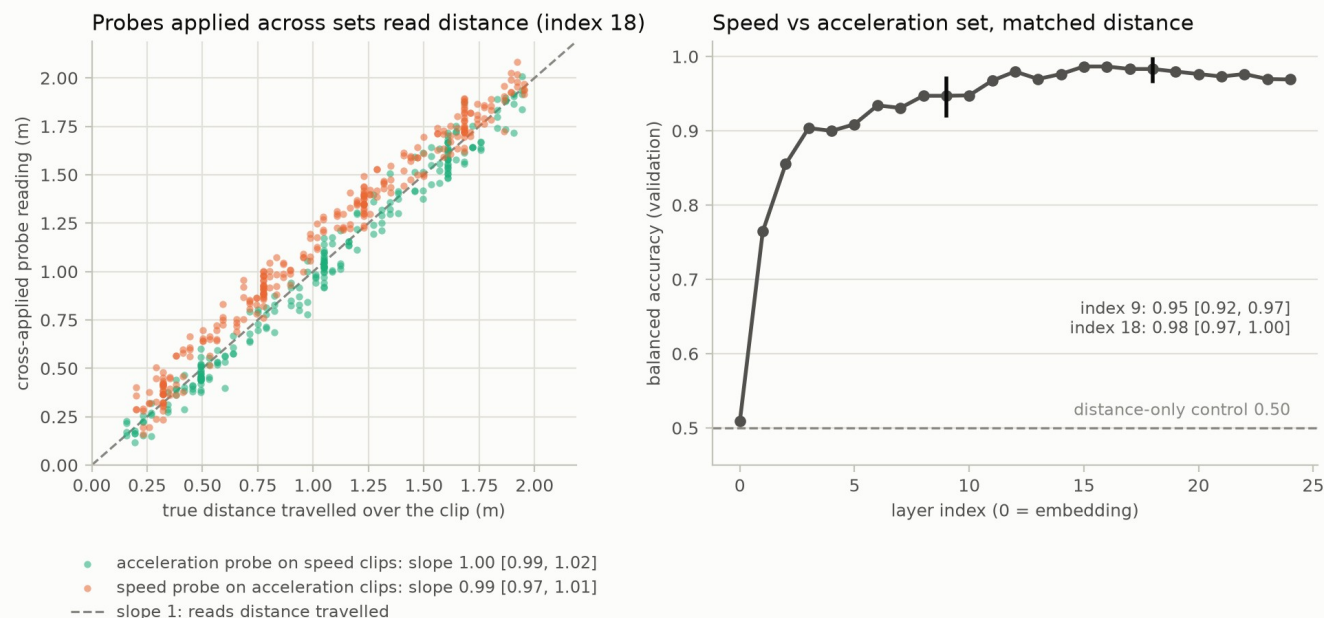
Same test clips, targets, layer and readouts for both methods

	Spline (manifold) steering	Multi-probe subspace steering
Strengths	No procedure-dependent count; a 6–8-dimensional curve; reaches held-out values; natural paths at the steering layer, especially on direction's loop	Needs no centroids; any target value; no curve to fit
Limitations	Needs labelled, balanced values; no extrapolation beyond the fitted range; “natural” is judged by our own readout, which is blurry for speed and acceleration	The count K depends on the ridge penalty; for speed and acceleration the edits are about twice as long (0.52–0.53 vs 0.26–0.31 × clip distance)
Failure cases	Exact interpolation (Goodfire-style) overfits: 1.3–1.5 × worse on held-out centroids	One probe fails even at the same layer; all K probes push activations off-distribution
Shared	Both fade downstream (0.07–0.14 at index 18); both use the same uniform token edit; at a linear readout they are equal by construction	

The probes read distance, yet the motion profiles stay separable

Left: speed and acceleration probes applied to the other set (index 18). Right: speed-set vs acceleration-set classifier at matched distance, by depth

The probes read distance travelled, yet the two motion profiles stay separable



Applied across sets, both linear probes read distance travelled: slopes 1.00 / 0.99; a probe that read acceleration itself would give slope 0 on constant-speed clips

Yet the model separates the motion profiles at matched distance: 0.95 / 0.98 at index 9 / 18; distance-only control 0.50

Robustness: direction transfers across motion types from index 9; clip flags don't drive errors beyond a $\sim 1^\circ$ excess for clipped clips; direction is near-orthogonal to speed and acceleration

What reproduced and what didn't

Our results against the papers' claims (physics paper: Joseph et al.; manifold paper: Goodfire / Wurgaft et al.)

Paper claim	Ours	Verdict
Speed and acceleration readable early (physics §5.3)	Index 1 of 24	Reproduced
Direction reliable only at the one-third emergence zone (§5.3, Fig. 2c)	R^2 0.85 at index 1	Differs
Per-patch direction goes local → global at one-third (App. C.5, Fig. 18)	From block 1	Differs
Direction needs tens to hundreds of dimensions (§7.2, Table 3)	6 rounds (10 dimensions) at index 9; depends on the ridge penalty	Differs (count depends on the ridge penalty)
Multi-probe steering works at the same layer; one probe doesn't (App. C.12)	0.92–0.95; one probe weak	Reproduced
Does the edit reach later layers? (not tested in the paper)	Fades within three blocks	New
Near-orthogonal probe subspaces (App. C.4)	Direction vs speed / acceleration at chance (probe weights)	In line (different pairs)
Concepts lie on curved manifolds (Goodfire)	Curved; direction a closed loop	Reproduced
Linear steering “teleports”; manifold steering stays natural (Goodfire §3.2)	At the steering layer only	Partly
Representation and behavior are isometric (Goodfire)	Holds, but no more than label order	Partly

Conclusions

- 1 Motion is readable after the first block,** direction included, and spread across the frame
slides [6](#), [7](#)
- 2 The nullspace count measures the procedure:** 7 to more than 150 with the ridge penalty; a nonlinear probe still reads it after linear erasure
slides [8](#), [9](#)
- 3 Steering edits work where they're made and fade within three blocks,** for probe and spline steering alike
slides [10](#), [11](#), [14](#)
- 4 The geometry is curved and low-dimensional;** following it keeps paths natural at the steering layer, but doesn't make edits propagate
slides [12](#), [13](#)

Caveat: speed and acceleration probes read distance travelled, yet the model separates the two motion profiles (slide [16](#))

Limitations and next experiments

Limitation or open question	Next experiment
Every method adds the same shift to every token	Token-structured edits (disk tokens, per time step) through the saved per-token cache; the time-structured edit is a first lead (+0.01–0.03)
One steering layer (9) and one readout layer (18)	Steer at several layers; read the model's own predictor forecast
Per-patch vectors are time-averaged	Per-time-step patch probes, closer to the paper's protocol
Speed and acceleration labels $\propto$ distance travelled	Clips that decouple them (e.g. accelerating from a non-zero start speed)
Linear erasure with one kernel family	Other nonlinear probes after erasure
Simpler stimuli, one checkpoint; parity with Meta's model taken from Hugging Face's conversion check	Other V-JEPA 2 sizes; our own parity run
Open: direction edits nudge the speed and acceleration readouts slightly more than random edits	Test the cross-talk mechanism

Thank you

Questions and discussion

Code and results github.com/shivansh052k/world-mechanics-take-home-shivansh-gupta

Contact shivanshgupta323@gmail.com

Appendix [supporting analyses on slides 21–28](#)

Appendix: supporting analyses

Each slide backs a claim in the main talk; titles link to the slide

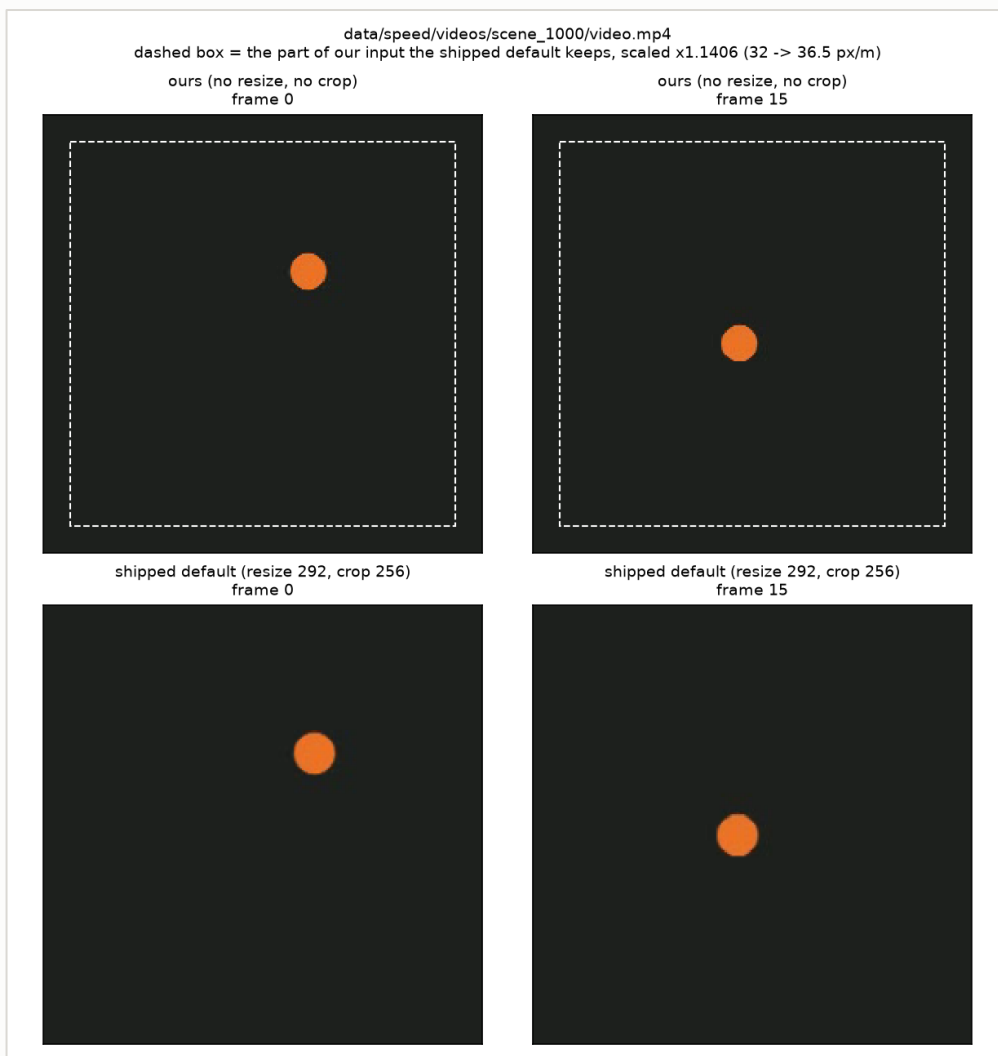
	Slide	Title	Supports
A1	22	Preprocessing: why resize and crop are off	slide 5
A2	23	Audit: label design and disk tracking	slide 4
A3	24	Direction transfers across motion types	slide 16
A4	25	Clip flags barely matter; slow motion is still read as a magnitude	slide 16
A5	26	Subspace overlap and steering cross-talk	slide 16
A6	27	Isometry against a label-order null	slide 12
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Preprocessing: why resize and crop are off

A1

One speed clip, first and last frame: our input (top) vs the checkpoint's shipped processor (bottom)



The shipped processor resizes to 292, then centre-crops 256. On our 256 px clips that rescales by $\times 1.14$ ($32 \rightarrow 36.5$ px per metre) and crops 18 px per side

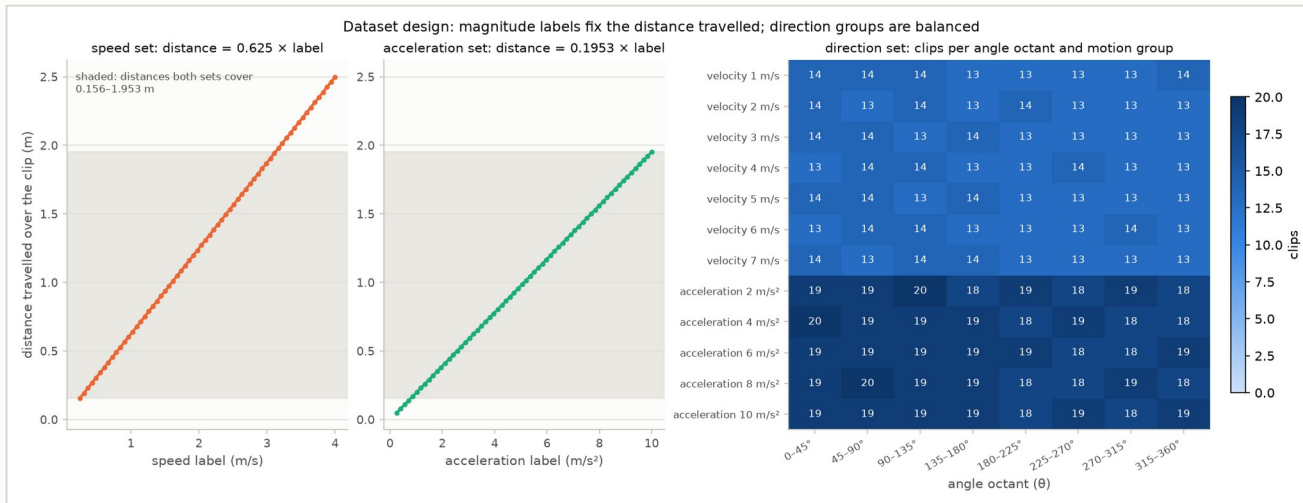
Ours: resize and crop off; rescale $1/255$ and ImageNet normalization kept. The output equals a manual normalization (max difference $1.65e-7$) and inverts to the decoded clip bit for bit

Our own hooks, not hidden_states: its last entry means different things across transformers versions

Audit: label design and disk tracking

A2

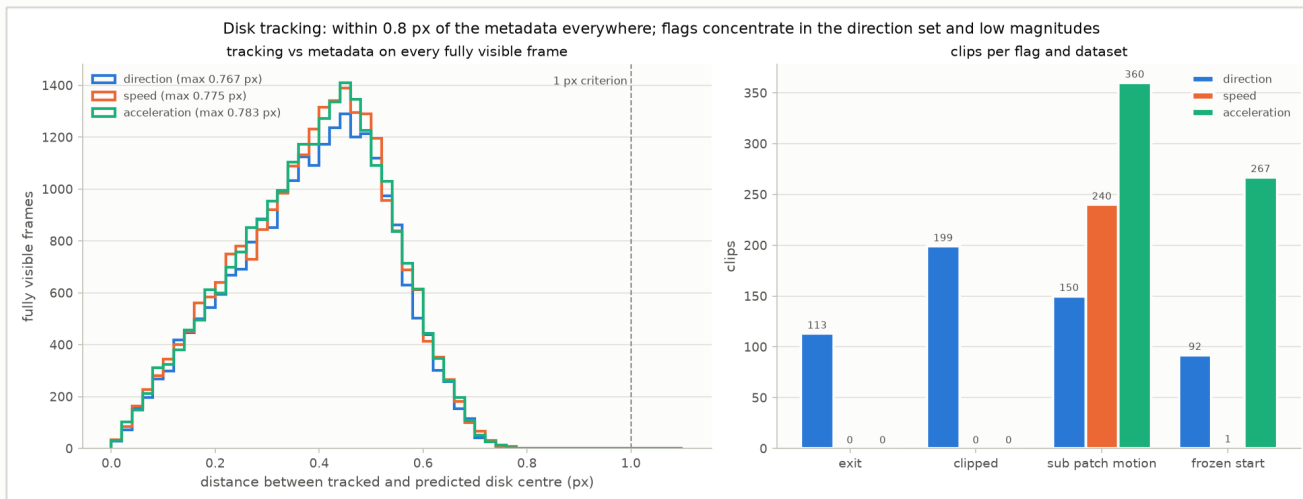
Top: labels fix distance travelled; direction groups balanced. Bottom: tracked disk vs metadata; clip flags per set



Labels fix distance travelled: distance = $0.625 \times \text{speed}$ and $0.1953 \times \text{acceleration}$; both sets cover 0.156–1.953 m. Direction's 12 motion groups are balanced across octants (13–20 clips per cell)

Tracking: the disk centre is within 0.783 px of the physics on every fully visible frame

Flags: exit 113 and clipped 199 (direction only); sub-patch motion 150 / 240 / 360; frozen start 92 / 1 / 267. All clips kept



Direction transfers across motion types

Across-type R²: direction probes fit on one motion type, scored on the other (validation / one-time test)

index	velocity → acceleration (val / test)	acceleration → velocity (val / test)
1	0.710 / 0.742 · partly shared	0.607 / 0.505 · partly shared
9	0.931 / 0.939 · shared, partly type-specific	0.962 / 0.948 · shared
18	0.972 / 0.969 · shared, partly type-specific	0.951 / 0.944 · shared, partly type-specific

Bands, fixed in advance: < 0.5 type-specific · 0.5–0.9 partly shared · ≥ 0.9 shared

All six readings hold on test

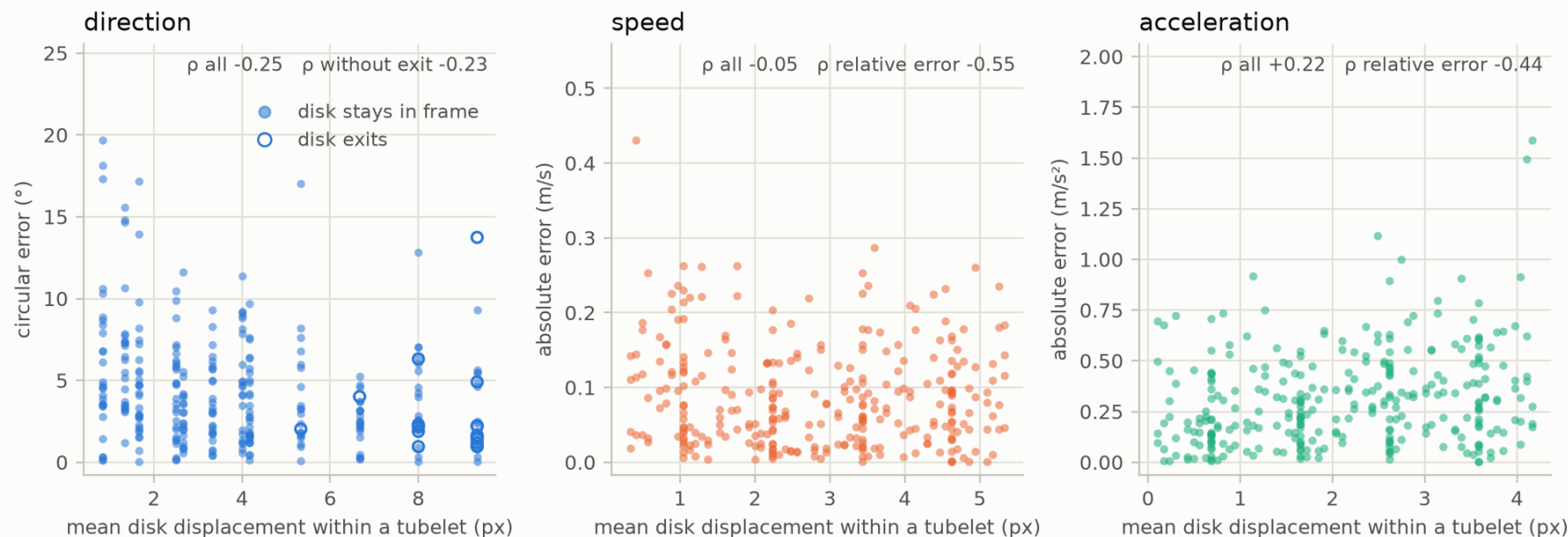
Within the distance range both types reach, the gaps at indices 1 and 9 disappear; index 18 acceleration → velocity keeps one (test 3.3° → 6.2°)

Clip flags barely matter; slow motion is still read as a magnitude

A4

Probe error vs mean disk displacement within a tubelet (validation clips, index 9); ρ is confounded with label value

Probe error vs per-tubelet motion (validation clips, index 9)



Each dot = one clip. ρ = Spearman correlation of error with displacement; displacement grows with the label, so ρ is confounded with label value.

Flags:

only the clipped-clip excess at index 18 holds on both splits (+1.24° validation, +0.72° test); exit and frozen-start effects flip between splits

Slow motion is read as a magnitude, not as “moved / didn’t move”:

inside the sub-patch range, readings still order the label on both splits (validation slope 0.58–0.98 at indices 1 / 9 / 18)

Relative error is 3–6 × higher

for sub-patch clips, which is confounded with label value

Subspace overlap and steering cross-talk

Index 9, common standardized space; subspaces fit on train only

pair	probe weights (1 / K – 1 probes)	patterns
speed–acceleration	overlap 0.36 / 0.37, 53° / 54°: aligned beyond chance	0.71, 33°: aligned
speed–direction	at chance (84–88°)	beyond chance but tiny (0.008, 83°)
acceleration–direction	at chance (84–89°)	at chance

Same readings in raw feature space. Direction's near-orthogonality is in line with the chance-level overlaps the paper reports between its probe subspaces

The readouts are valid on the other sets' clips (R^2 0.93–0.98), so cross-talk is interpretable

Cross-talk beyond a random edit (index 18): direction steering moves the speed readout +0.012 m/s [0.005, 0.019] and the acceleration readout +0.055 m/s² [0.031, 0.080]; speed or acceleration steering leaves direction alone

Isometry against a label-order null

Behavior readout at index 18; activation and behavior curves from disjoint halves of the training clips

	direction	speed	acceleration
r (activation geodesic, behavior geodesic) / r (label, behavior)	0.990 / 0.991	0.968 / 0.982	0.890 / 0.995
geodesic – label [95 % CI]	–0.001 [–0.004, 0.002]	–0.015 [–0.024, –0.006]	–0.105 [–0.131, –0.074]
local agreement per unit label, r [95 % CI] (split-half ceiling)	0.09 [–0.19, 0.40] (0.38)	0.59 [0.27, 0.77] (0.62)	0.41 [–0.10, 0.65] (0.64)

Goodfire's pattern holds: the geodesics correlate at 0.89–0.99

But label distance alone does as well or better: on a one-dimensional variable, both geometries follow label order

Locally, only speed's two curves agree beyond the label. Caveat: equal-width value bins make behavior distances inherit label spacing partly by construction

Spline vs multi-probe: full comparison

A7

Gain = output-space gain (1 = full effect); naturalness = Hellinger distance to the behavior curve, below 0 = more natural

Block	Quantity	Direction	Speed	Acceleration
Geometry	held-out centroid error, reference → curve	144.0 → 56.4 (ellipse)	86.7 → 45.5 (line)	85.5 → 50.5 (line)
	reference – curve [95 % CI]	87.6 [70.8, 105.1]	41.3 [30.9, 53.6]	35.0 [21.4, 53.5]
	curvature share of the gain	—	87 %	77 %
	curve PCA dimensions / nullspace dimensions (K × outputs)	8 / 12	6 / 7	6 / 7
	participation ratio of the curve	3.14	1.59	1.52
Steering	edit size ÷ clip distance: spline / covariance / K – 1 probes	0.58 / 0.47 / 0.52	0.31 / 0.23 / 0.52	0.26 / 0.23 / 0.53
	spline vs covariance edit difference (median)	0.75	0.76	0.66
	gain index 9 / 18 / mean 10–18: spline	0.986 / 0.069 / 0.269	0.973 / 0.115 / 0.355	0.956 / 0.118 / 0.306
	· covariance line	0.986 / 0.069 / 0.268	0.977 / 0.119 / 0.357	0.957 / 0.119 / 0.308
	· K – 1 probes	0.980 / 0.068 / 0.271	1.011 / 0.128 / 0.355	0.993 / 0.136 / 0.317
	· random (floor)	0.001 / 0.001 / 0.000	0.005 / 0.001 / 0.003	–0.006 / –0.004 / 0.000
Naturalness	midpoint excess, spline – chord, index 9 [CI]	–0.124 [–0.138, –0.110]	–0.046 [–0.055, –0.037]	–0.015 [–0.025, –0.004]
	same, index 18 [CI]	0.000 [–0.004, 0.004]	–0.001 [–0.003, 0.000]	–0.001 [–0.004, 0.001]
Held-out	spline gain index 9, seen / unseen targets	0.983 / 0.989	0.971 / 0.974	0.987 / 0.942
Timing	time-structured – uniform, mean gain 10–18 [CI]	0.030 [0.028, 0.032]	0.013 [0.012, 0.015]	0.017 [0.015, 0.019]
	time-reversed – uniform [CI]	–0.026 [–0.028, –0.024]	–0.007 [–0.008, –0.006]	–0.007 [–0.009, –0.005]